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#include<stdio.h>

#define MAX 20

struct Process {
    int pid, at, bt, ct, tat, wt, type;
};

int main() {
    printf("Neha Kumari Sah (1WA24CS185)\n");

    struct Process p[MAX], sq[MAX], uq[MAX];

    int n, i, sq_count = 0, uq_count = 0, time = 0;

    printf("Enter number of processes: ");

    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        printf("\nProcess %d\n", i + 1);

        p[i].pid = i + 1;

        printf("Arrival Time: ");

        scanf("%d", &p[i].at);

        printf("Burst Time: ");

        scanf("%d", &p[i].bt);

        printf("Type (1=System,2=User): ");

        scanf("%d", &p[i].type);

        if (p[i].type == 1)
            sq[sq_count++] = p[i];
        else
            uq[uq_count++] = p[i];
    }

    for (i = 0; i < sq_count; i++) {

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    if (time < sq[i].at)
        time = sq[i].at;
    time += sq[i].bt;
    sq[i].ct = time;
}

for (i = 0; i < uq_count; i++) {
    if (time < uq[i].at)
        time = uq[i].at;
    time += uq[i].bt;
    uq[i].ct = time;
}

float total_tat = 0, total_wt = 0;
printf("\n---System Queue---\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");
for (i = 0; i < sq_count; i++) {
    sq[i].tat = sq[i].ct - sq[i].at;
    sq[i].wt = sq[i].tat - sq[i].bt;
    total_tat += sq[i].tat;
    total_wt += sq[i].wt;
    printf("%d\t%d\t%d\t%d\t%d\t%d\n",
           sq[i].pid, sq[i].at, sq[i].bt,
           sq[i].ct, sq[i].tat, sq[i].wt);
}

printf("\n---User Queue---\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");
for (i = 0; i < uq_count; i++) {

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        uq[i].tat = uq[i].ct - uq[i].at;

        uq[i].wt = uq[i].tat - uq[i].bt;

        total_tat += uq[i].tat;

        total_wt += uq[i].wt;

        printf("%d\t%d\t%d\t%d\t%d\t%d\n",

                uq[i].pid, uq[i].at, uq[i].bt,

                uq[i].ct, uq[i].tat, uq[i].wt);

    }

    printf("\nAverage Turnaround Time=%.2f", total_tat / n);

    printf("\nAverage Waiting Time=%.2f\n", total_wt / n);

    return 0;

}

```

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C:\Users\sahne\OneDrive\De
Neha Kumari Sah (1WA24CS185)
Enter number of processes: 4

Process 1
Arrival Time: 0
Burst Time: 4
Type (1=System,2=User): 1

Process 2
Arrival Time: 0
Burst Time: 3
Type (1=System,2=User): 1

Process 3
Arrival Time: 0
Burst Time: 8
Type (1=System,2=User): 2

Process 4
Arrival Time: 10
Burst Time: 5
Type (1=System,2=User): 1

---System Queue---
PID   AT    BT    CT    TAT   WT
1      0     4     4     4     0
2      0     3     7     7     4
4      10    5     15    5     0

---User Queue---
PID   AT    BT    CT    TAT   WT
3      0     8    23    23    15

Average Turnaround Time=9.75
Average Waiting Time=4.75

Process returned 0 (0x0)   execution time : 31.266 s
Press any key to continue.

```

